

Jeena Smart Document Parser (JSDP)

Project Proposal for Senior Developer Review

1. Project Overview

The Jeena Smart Document Parser is an AI-assisted web application designed to reduce manual data entry in logistics operations. Employees upload logistics documents such as Commercial Invoices, Packing Lists, Air Waybills, Bills of Lading and Delivery Orders. The system extracts important shipment information, validates it, allows human review, and stores structured data for further business processes.

2. Business Problem

Logistics teams spend significant time manually reading documents and entering shipment details into internal systems. Manual entry is slow, repetitive and error-prone. The proposed system automates document understanding while keeping a human-in-the-loop review step.

3. Objectives

- Upload logistics documents
- Detect document type
- Extract structured fields
- Validate mandatory fields
- Display confidence scores
- Allow manual correction
- Store structured information in PostgreSQL
- Search and retrieve processed documents
- Build a dashboard with processing statistics

4. Suggested Architecture

Angular frontend communicates with Django REST Framework APIs. The backend performs authentication, document management, OCR and AI extraction, and persists data in PostgreSQL.

5. Technology Stack

Frontend: - Angular - Angular Material - TypeScript - HTML/CSS Backend: - Python - Django - Django REST Framework - JWT Authentication Database: - PostgreSQL Document Processing: - pdfplumber - Tesseract OCR - OpenCV AI: - Gemini API or OpenAI API for structured extraction Developer Tools: - Git - GitHub - Postman - VS Code Future: - Docker - Celery - Redis - Nginx - Azure/AWS Deployment

6. Core Modules

1. Authentication 2. Dashboard 3. Document Upload 4. Document Classification 5. OCR/Text Extraction 6. AI Field Extraction 7. Validation & Review 8. Search & Filters 9. Analytics Dashboard 10. Audit Logs

7. Example Extracted Fields

Invoice Number, Invoice Date, Customer Name, Consignee, Shipper, AWB Number, Bill of Lading Number, Gross Weight, Net Weight, Currency, Invoice Value, Country, HS Code, Incoterm.

8. Database (High Level)

Users Documents ExtractedFields Customers AuditLogs

9. Development Roadmap

Phase 1: Project setup, authentication, upload. Phase 2: OCR and text extraction. Phase 3: AI field extraction. Phase 4: Database integration. Phase 5: Search and dashboard. Phase 6: Testing, optimization and deployment.

10. Expected Benefits

• Reduced manual effort • Faster document processing • Lower human error • Better searchability • Improved operational efficiency • Foundation for future logistics AI automation

11. Future Enhancements

Email attachment processing, ERP integration, shipment creation, multilingual support, anomaly detection, conversational search over processed documents.